

CTO Governance — Logic Map

Tier A, lite-Midnight-reuse design (v4 — design settled) · real ZK voting, zero Midnight wallet extension, creator gets one telegraphed chance to cancel, creator-cap evasion closed · grounded against direct source reads, not recollection

● Already built

● Net-new / gap

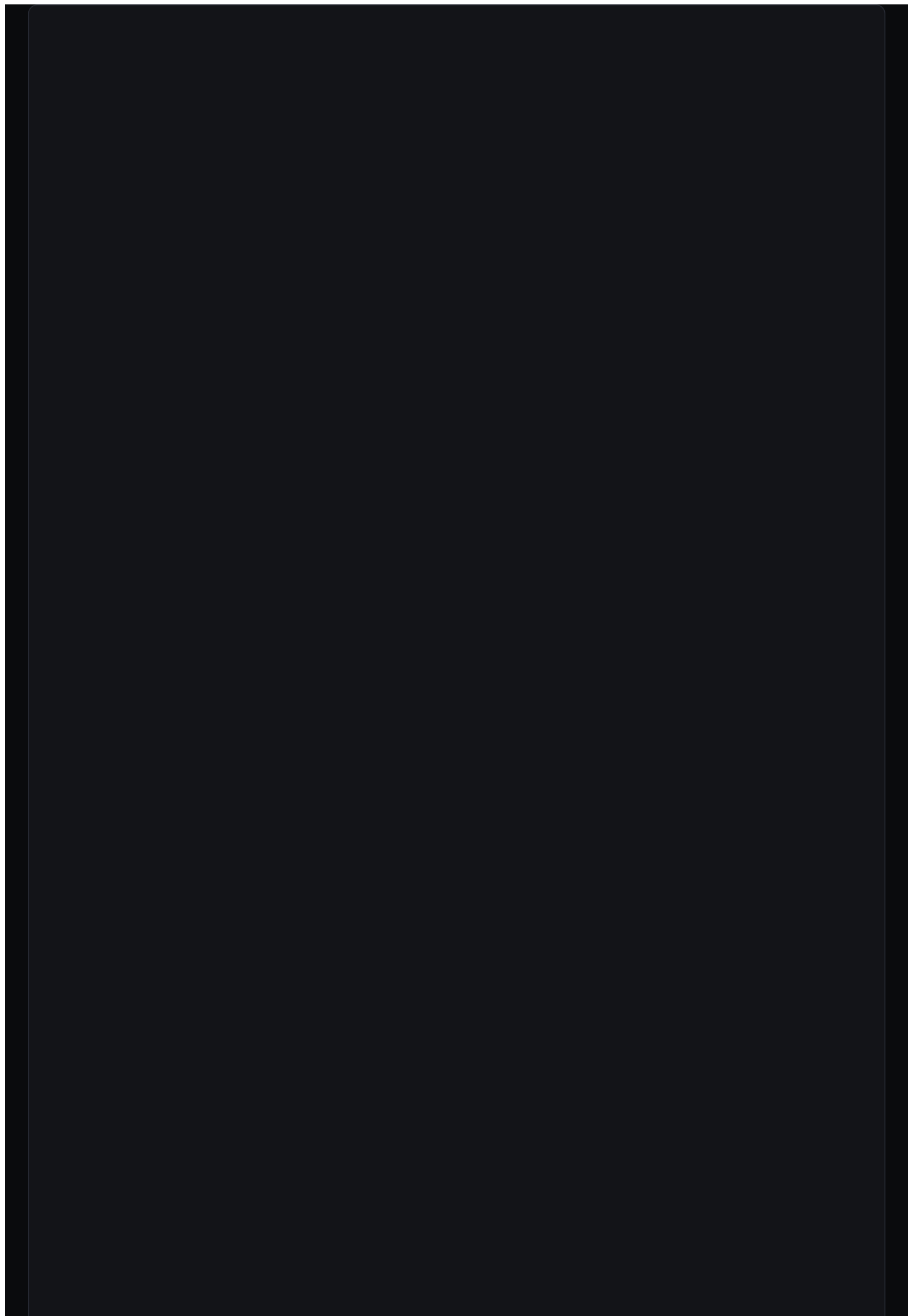
● Needs a decision

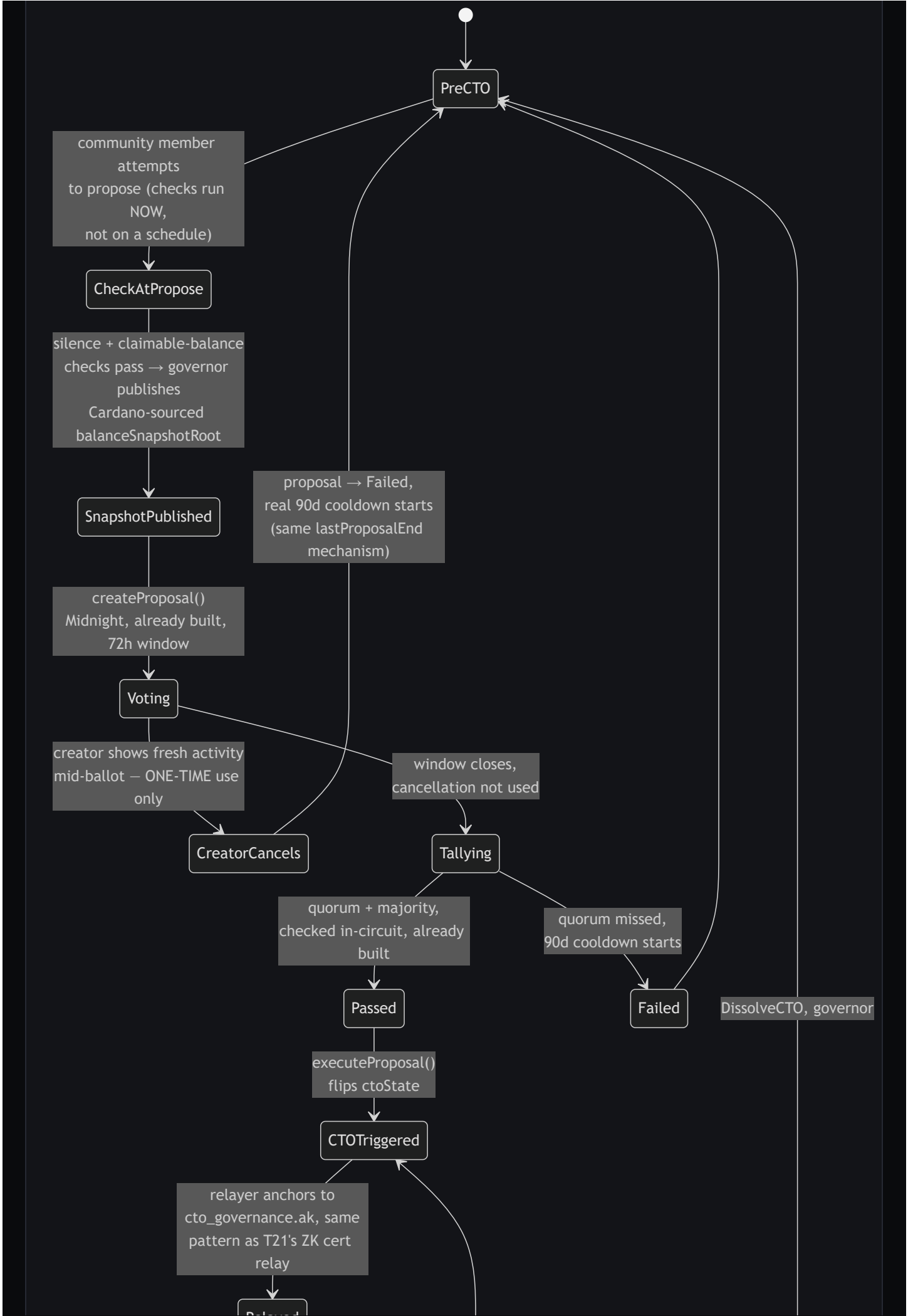
Revision history

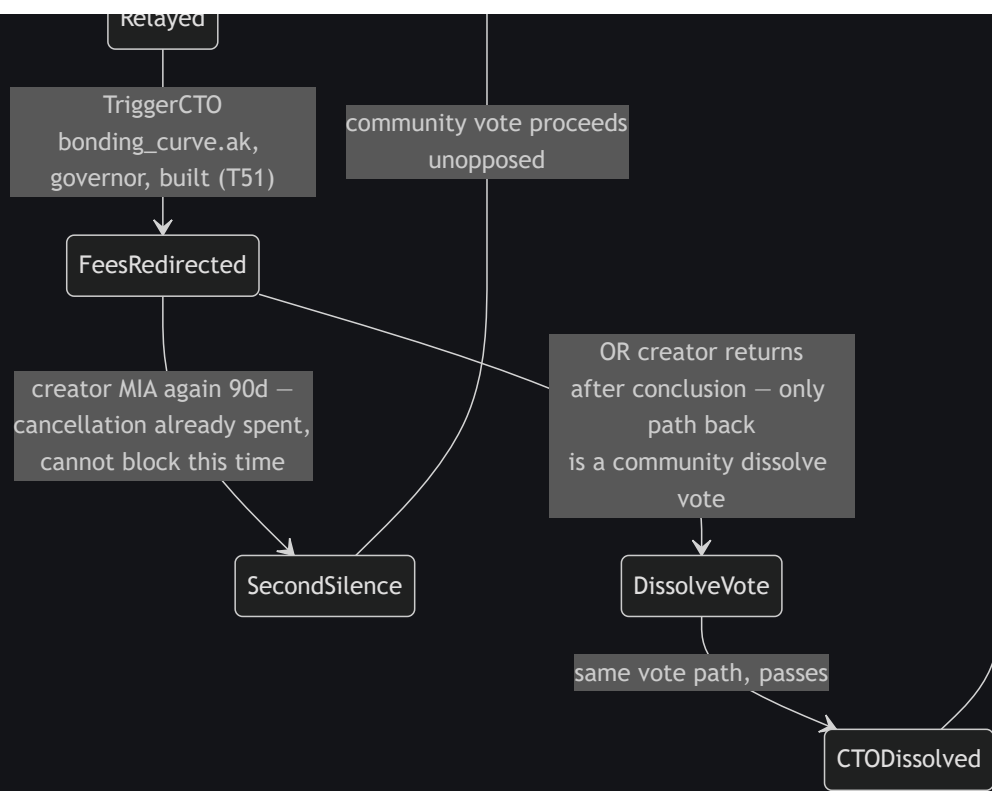
v1 assumed a bespoke Cardano-only off-chain tally, no ZK, several open trust questions. **v2** found `cto_governance.compact`'s real silence-tracking, gated proposal creation, and audited ZK voting already built — redesigned around reusing it with a Cardano-sourced balance snapshot. **v3** resolved the wallet-UX question via client-side proof generation from a derived CIP-8 identity, and added the creator's one-time mid-ballot cancellation path. **v4** (this version) closes the last two open questions: "creator active again" reuses the same activity feed as the silence check, and linked-wallet detection is reframed around its real target (creator-cap evasion, not general sybil farming) and resolved using two checks already built and proven for DarkVeil. No open questions remain — next step is implementation.

1 State machine

Two proposal types actually change state. The other three (FundAllocation / DexMigration / WhitelistUpdate) reuse the same machinery but never touch `ctoState` — omitted below for clarity. Solid boxes below split by which side of the bridge each step runs on.







The cancel-once-then-no-mercy shape is deliberate, not a loophole to close later — see the callout below on documenting it as a fair, telegraphed consequence rather than a takeover.

2 Triggers

What actually has to be true before each proposal type can open — plus the two new voter-eligibility rules from this round (min \$50, one-vote-per-wallet, linked-wallet check).

PROPOSAL TYPE	PRECONDITION	VERIFIABLE HOW	STATUS
SilenceLockTrigger	No ClaimCreatorFees for 90+ consecutive days <i>and</i> no public post on a verified project channel for 90+ days	Fee-claim half: real, derivable today from the trade-history tables. Social half: real, low-cost checks now confirmed — Discord/Telegram Bot APIs are free; X/Twitter is pay-per-use (\$0.005/read) since Feb 2026, ~\$0.15/mo per launch at daily polling frequency.	● checkSilenceLock() built ● Activity feed is new
DissolveCTOProposal	CTO currently triggered; creator demonstrably active again	No defined "creator is active again" check exists — presumably the inverse of the silence check, but never specified.	● Needs a decision
FundAllocation / DexMigration / WhitelistUpdate	30 days minimum post-graduation; 90-day cooldown after any prior vote	<code>createProposal()</code> already checks <code>minPostGradDelay</code> and <code>cooldown</code> in-circuit.	● Built
Voter eligibility (all types)	Holds $\geq$ \$50 USD of the launch token at snapshot time; one vote per wallet; not part of a confirmed linked-wallet cluster	\$50 filter: applied when the governor builds the snapshot — ineligible wallets simply get no leaf (same pattern as DarkVeil's NIGHT-bond filter). Price comes from generalizing the already-real Minswap pool-price + Orcfax ADA-USD chain. One-vote: nullifier, in-circuit, already built. Linked-wallet: no existing mechanism — proposed reuse of <code>nhop_challenge.ak</code> 's bonded-challenge idiom, applied before the snapshot is published.	● One-vote built ● \$50 filter + sybil check are new

3 Stage-by-stage: what's needed

Walking the `SilenceLockTrigger` → `CTOTriggered` path. Far more of this turned out to be already built and audited than v1 of this doc showed.

0 · Activity feed → `updateCreatorActivity()`

NEW, BUT THE HOOK ALREADY EXISTS

`updateCreatorActivity(timestamp, hasClaimableBalance)` is a real, governor-only Midnight circuit that already exists specifically to be fed by off-chain monitoring — its own comment says it fires "when the creator claims fees OR when a social post is detected." Fee-claim events are already decoded by the trade-history reader. Social-post detection needs a new poller: Discord/Telegram bots (free) and/or X's pay-per-use API (~\$0.15/mo/launch at daily frequency) feeding the `max(last activity)` across a launch's registered channels into this call.

New: a scheduled job calling an already-built circuit. Not a new contract.

1 · Silence check

FULLY BUILT

`checkSilenceLock(currentTimestamp)` computes `elapsed >= silenceThreshold` on-chain, in Compact, today. Nothing to build.

2 · Balance snapshot (Cardano-sourced)

NEW DATA SOURCE, EXISTING MECHANISM

`updateBalanceSnapshot(newRoot)` and `balanceSnapshotRoot` already exist — a governor-only circuit publishing a Merkle root of `(voterKey, balance)` leaves, pinned per-proposal so a later re-publish can't retroactively change what an in-flight vote proves. For B/C this reflects Midnight-native balances; for Tier A it needs a new off-chain builder that reads real Cardano balances via Blockfrost, filters to $\geq$ \$50 USD-equivalent (reusing the generalized Minswap pool-price pattern), excludes any confirmed sybil cluster, and publishes the resulting root.

New: a Cardano-balance snapshot builder script. The contract-side circuit needs no changes.

3 · Proposal creation, 72h window

FULLY BUILT

`createProposal()` already enforces: a real snapshot must exist, `minPostGradDelay` has passed, and the cooldown since the last proposal has elapsed. `ballotDuration` is a real 259,200-second (72h) constant. No new WordPress-side ballot-state table needed — the ballot genuinely lives on Midnight.

4 · Vote collection + tally

FULLY BUILT, ZK-PRIVATE, AUDITED

`castVote()` already does everything: nullifier-based one-vote-per-wallet, vote weight proven via Merkle membership against the pinned snapshot (not a self-declared amount), creator weight capped in-circuit at 2% with the real uncapped amount still tracked for audit (T35), and the vote itself is private — which way you voted isn't linkable to your identity. Multiple "Security audit fix" comments in the source suggest this already went through a real review pass.

4a · Creator mid-ballot cancellation

NEW CIRCUIT + NEW STATE FIELD

Confirmed real: `cooldownDuration = 7,776,000` seconds — exactly 90 days, already a live contract constant. A cancellation just needs to reach the same `lastProposalEnd`-setting path `finalizeProposal` already uses, early. Needs: a new circuit (structurally parallel to `finalizeProposal`, but callable while `state == Active` and before `endTimestamp`) checking caller identity == `creatorKey` + fresh activity, setting the proposal to `Failed` and triggering the existing cooldown — no new enum variant required. Plus one genuinely new field: a per-launch `creatorHasCancelledOnce` flag, since nothing today tracks whether the one-time privilege has been spent.

4b · Documentation: not a hostile takeover

DOCS, NOT CODE

The two-strike shape (one free cancellation while genuinely active, no mercy on a second 90-day silence) is fair specifically *because* it's telegraphed in advance. Needs writing into CLAUDE.md's CTO Governance section and any public-facing docs/whitepaper language, not left implicit in contract comments — the whole point is that neither creators nor the community should discover these rules only when they're invoked.

5 · Bridge to Cardano

NEW, BUT A PROVEN PATTERN

`cto_governance.compact`'s own header is explicit: it "cannot call any other PSM's `triggerCTO` directly" and each dependent contract's trigger "must be called as a separate, off-chain-orchestrated transaction." Once a Midnight proposal executes, a relay needs to anchor the result to `cto_governance.ak` on Cardano — the exact same shape as T21's real ZK Fair Launch Certificate relay, not a new design pattern.

6 · Consequences fire

BUILT, NEVER EXERCISED ON PREPROD

Once anchored, a governor-signed `TriggerCTO` call on `bonding_curve.ak` (and the `lp_escrow.ak` equivalent) redirects fees to the community wallet — written since T51, just never run for real end-to-end. Whether `vesting.ak` has the matching unvested-token-freeze hook still needs checking.

7 · Dissolve path

SAME MECHANISM, REVERSED

Identical vote → execute → relay → trigger flow with `DissolveCTO`, flips `ctoState` back and reverses the fee redirect on both Cardano contracts.

4 Resolved this round

Lite-wallet UX

CONFIRMED BUILDABLE

No Midnight wallet extension needed. One-time CIP-8 signature (already-built) derives a recoverable Midnight identity (matching the already-decided "derive, don't generate" principle); `witnesses.ts`'s own `getUserSecret: () => [undefined, userSk]` confirms the witness layer has zero coupling to a connected wallet — it's just a value. Proof generation should happen client-side (matching the file's own "generated client-side, never shared on-chain" convention) so the platform never sees how a wallet voted; the platform's own Midnight wallet submits and pays DUST, same sponsorship model as DarkVeil.

Snapshot/check timing

CONFIRMED

Checks run at the moment a community member attempts to propose, not on a schedule — `createProposal()` already gates this in-circuit. A background poller can surface eligibility as a UI hint, but isn't the authoritative gate.

"Creator is active again"

CONFIRMED — REUSES STAGE 0

Same activity feed as the silence check, read in the opposite direction: a fresh `ClaimCreatorFees` or a fresh verified-channel post counts as "active again." No new monitoring — gates the standalone `DissolveCTOProposal` path (creator returns after a CTO already concluded). The mid-ballot cancellation path (4a) uses the same feed for its own, earlier check.

Sybil/linked-wallet detection

REFRAMED + RESOLVED

The real exploit is narrower than general sybil resistance. Vote weight is a wallet's real balance, and balance is additive — splitting tokens across wallets doesn't gain a community member any extra total weight, unlike DarkVeil's fixed-allocation registration problem. The actual risk is `castVote`'s creator-cap check being a single identity comparison (`voter.bytes == creatorKey`): a second, unregistered wallet the creator secretly controls votes as an ordinary community member with no 2% cap applied.

Primary defense (automatic, no bond): when building the Cardano balance snapshot, run the launch's candidate wallets against the creator's registered wallet using the two checks already built and proven for the identical question in DarkVeil registration — `checkStakeKeyMatch` (a base address's stake credential is in its own bytes, no signature needed) and `checkNoDirectAdaFlow` (direct ADA transfer from the creator's wallet within a lookback window). A match just doesn't get a leaf — same silent-exclusion pattern used everywhere else. Zero new detection code.

Secondary defense (bonded challenge, for links the automatic check can't see): reuse `nhop_challenge.ak`'s pattern as-is, including the existing `NHOP_CHALLENGE_BOND = 25 ADA` constant. Window sits between snapshot publication and ballot start, not after votes are cast — contesting a wallet post-vote would mean recomputing quorum/majority after the fact.